Electronic Signature Information

Name	DOC2867146
Revision	2
Type	Software Architecture Design/ Software Detail Design/ Software Units/ PDD
Title	MR Cad-2024 SWADD
Reason For Change	Program M3
Originator	212070039_andrew_j._byshenk
Release Date	11/01/2024 11:16:20 PM
Obsolete Date	

File Name	File Description	File Size (Bytes)
GEHC_MR_Cad-2024_SwArchitecture.docx	Updated Host Platform SSDD reference.	3979162

Route	Signer	Function	Status	Comments	Completion Date
R-11237239	212311476_madhur__josan		Approve	Approved as a Task Assignee	1 Nov 2024 15:53:19 GMT

Periodic Review

There are no signatures or routes related to this business object.

Obsolesence Approval

There are no signatures or routes related to this business object.

* Printed versions are For Reference Only *
+ Indicates a task was reassigned from an original assignee

GE HealthCare

MR Cadence-2024
Software Architecture Design Description

Contents

Topic	Page
1. Document Overview	**3**
1.1. Purpose	3
1.2. Scope	3
1.3. References	4
1.4. Glossary	6
1.5. Ownership	7
1.6. Authorization	7
1.7. History	7
2. Software Overview	**9**
2.1. Context	9
2.2. Uses	10
2.3. Layering	11
2.4. Subsystems	12
2.5. Error Handling and Recovery	18
2.6. Risk Control	18
2.7. Legacy Reuse	19
2.8. Configurability	19
2.9. Off-the-Shelf Components	20
2.10. Extensibility	20
3. Environment	**21**
3.1. Hardware	21
3.2. Operating Software	22
3.3. Platforms	24
3.4. Libraries	24
4. Structure	**24**
4.1. Host Software	25
4.2. Host Platform	27
4.3. Visualization Applications	28
4.4. MR Reconstruction	29
4.5. Pulse Sequence Software	34
4.6. MR Firmware	35
4.7. MR Service Software	37
5. Interfaces	**39**
5.1. User Interfaces	39

*** Printed copies are uncontrolled unless otherwise identified ***
Before using this document, consult MyWorkshop for the latest revision.

5.2. External Interfaces 42
5.3. Hardware Interfaces 43
5.4. Internal Interfaces 44
6. Functions 53
6.1. Primary Processing 54
6.2. Secondary Processing 62
6.3. Error Handling and Recovery 65
6.4. Internationalization and Localization 66
7. Data 67
7.1. Static Data 67
7.2. Dynamic Data 67
7.3. Acquisition Memory 67
7.4. Session Memory 67
7.5. Data Management 67
8. Safety Segregation 68
8.1. RF Power Protection 69
8.2. Image Flip Protection 70
8.3. Patient Identification Protection 71
9. Generation 72
9.1. Tools 72
9.2. Directories and Files 73
9.3. SOUP 73
10. Traceability 73

*** Printed copies are uncontrolled unless otherwise identified ***
Before using this document, consult MyWorkshop for the latest revision

Released

1. DOCUMENT OVERVIEW

This document describes the architecture of the software for MR-only products (not PETMR) supporting Cadence-2024 in the MR modality of GE HealthCare. The software supports functionalities in the GEHC MR modality scanner systems as described in the MR System Design Description [12] and further clarified in the Software Development Plan [18]. In this document, the scanner system is identified as the Scanner and the supporting software is identified as the Software.

Note: Cadence-2024 serves the following program releases: Capri-B, Tang-B, a 7T program, and Kun Peng.

1.1. Purpose

This document provides a high-level architectural view of the Software, describing its interactions with users and non-software components of the Scanner, its internal organization, its operations, and the information it consumes and generates.

The internal design details of the subsystems are expected to evolve without impacting this document. Detailed designs of the Software subsystems, including their contained software items and internal interfaces are described in the next lower level of documentation.

1.2. Scope

This description covers the following aspects of the Software.

- The Software Overview section (2) describes the context in which the Software operates, the uses that the Software fulfills, the layering of the Software deployment, the top-level organization of the Software subsystems, the Software's high-level error handling and recovery mechanisms, the high-level risk control measures applied in the Software, and how this Software fits within the continuum of GEHC software in the past (legacy) and future (extensibility).
- The Environment section (3) describes the hardware and operating systems on which the Software runs, the engines used to execute code written in specific languages, and the platforms used across multiple subsystems.
- The Structure section (3.4.2) describes the primary components of the Software subsystems, the responsibilities of the Software subsystems, and the relationships of the Software subsystems with other Scanner components (HW and SW).
- The Interfaces section (5) describes how the Software subsystems interact with other Scanner elements and components outside the Scanner (externals, users).
- The Functions section (6) describes the primary and secondary uses supported by the Software, specifically identifying the participating subsystems, their interactions, and the major process and data flows.
- The Data section (7) describes various ways that the Software handles information that it uses or produces.
- The Safety Segregation section (8) describes the practices used to separate software modules so that components supporting safety characteristics are not compromised by those with no safety associations.

* Printed copies are uncontrolled unless otherwise identified *
Before using this document, consult MyWorkshop for the latest revision.

Released

- The Generation section (9) describes at a high level how the Software is produced, including identifying the directory and file organization, the tools used for building, and the third-party software utilized.

The Software supports a variety of Scanner configurations comprised of varying hardware components and applications. Support for specific collections of hardware and applications is enabled by declaration, installation selections, or the presence/absence of the hardware components. Valid hardware configurations are described in the detailed designs for the associated hardware subsystems.

Some features are not accessible on specific Scanner configurations, although the software support may be present; refer to the product line specification sheets for compatibility details.

Software requirements flow down from the system-level requirements and then flow down to the respective software items appropriately. (Refer to the appropriate software requirements documents.)

1.3. References

The following documents include information relevant to the understanding of this document's content.

[1]	GEHC Design Controls Procedure	DOC0268468
[2]	GEHC Design Controls Design Input Work Instruction	DOC0505546
[3]	GEHC Design Controls Design Output Work Instruction	DOC0505548
[4]	GEHC Design Controls Design Review Work Instruction	DOC0505552
[5]	GEHC Software Development Lifecycle Work Instruction	DOC1617896
[6]	Coreload Design History File Index	DOC2234468
[7]	Volume Share 7 Design History File Index	DOC1145961
[8]	Cyber Security Design History File Index	DOC2196823
[9]	Engineering CSV Tracker Waukesha MR	DOC0630293
[10]	MR Glossary	DOC0706894
[11]	MR Family RAC	DOC0457783
[12]	MR System Design Document	DOC0700133
[13]	MR Cause Mitigation Table (CMT)	DOC2324818
[14]	Capri DHF	BKS23313
[15]	Tang DHF	BKS23365
[16]	7T DHF	BKS21884
[17]	Kun Peng DHF	BKS23355
[18]	MR Cad-2023H2 Software Development Plan (SDP)	DOC2858495

*** Printed copies are uncontrolled unless otherwise identified ***
Before using this document, consult MyWorkshop for the latest revision.

[19]	MR Cad-2023H2 Software Structure and Safety Classification	DOC2867148
[20]	Regulatory FRS	See DHFs [14]/[15]/[16]/[17]
[21]	MR Host Subsystem FMEA	DOC2644767
[22]	MR Visualization Subsystem FMEA	DOC1519235
[23]	MR Reconstruction Subsystem FMEA	DOC1740093
[24]	PSD Subsystem FMEA	DOC1143808
[25]	MR Firmware Subsystem FMEA	DOC2320526
[26]	MR System Tools Subsystem FMEA	DOC1165663
[27]	MR Host Tools Subsystem FMEA	DOC1749344
[28]	Host Operating System SOUPRS	DOC3004080
[29]	VRE OS SW SOUPRS	DOC2705987
[30]	VxWorks OS SOUPRS	DOC2213449
[31]	NIOS II SOUPRS	DOC2213255
[32]	ACE TAO OpenORB SOUPRS	DOC1437460
[33]	Firmware ACE SOUPRS	DOC2212713
[34]	FFTW VIZ SOUPRS	DOC1433952
[35]	MR Host Software SOUPRS	DOC1032580
[36]	MR Recon Libraries SOUPRS	DOC1796149
[37]	MR Host Software SSDD	DOC2659998
[38]	MR Host Platform SSDD	DOC3062901
[39]	MR Visualization SSDD	DOC1897083
[40]	MR Reconstruction SSDD	DOC1896329
[41]	MR Prescan SSDD	DOC2836980
[42]	Pulse Sequence SSDD	DOC1542468
[43]	MR Firmware SSDD	DOC2945066
[44]	MR Host Tools SSDD	DOC2264318
[45]	MR System Tools SSDD	DOC2857675
[46]	Magnet Monitor SWDD	DOC2945076
[47]	Image Quality Engine SWADD	DOC2838292

All identifiers in this section are from MyWorkshop unless specified otherwise.

*** Printed copies are uncontrolled unless otherwise identified ***
Before using this document, consult MyWorkshop for the latest revision.

1.4. Glossary

The following terms have specific meaning in this document.

Table 1-1: Glossary

Term	Definition
ADS	Acquisition Data Server
Safety Class C	A classification for software items with failure modes that can result in Hazards with Critical or Catastrophic Severity. See GEHC SDLC [5] and section 8 of this document for more details.
CORBA	Common Object Request Broker
CSV	Computerized System Validation
DAB	Data Acquisition Board
FE	Field Engineer or Field Service Engineer
GEOCOMM	GE Object communication
GOC	Global Operator Console
HLA	High Level Access
i18n	Internationalization
ICN	Image Compute Node
IQE	Image Quality Engine
IROC	in-room operator console
MPPS	Modality Performed Procedure Step
OTS	Off-the-Shelf
PACS	Picture archive system
PCI	Peripheral Component Interconnect
PCIe	PCI Express
PEF	Platform Execution Framework
PSD	Pulse Sequence Description
PSI	Pulse Sequence Interface
PSS	Pulse Sequence Software
RIaS	Remote Installer at Site
ROI	Region of Interest
RSvP	Remote Service Virtual Platform
RTP	Real-time processing
SAT Band	Saturation Band
SOUP	Software of Unknown Pedigree
SSC	Scan and Sequence Control
SVAT	System Validation and Test
SWPE	Software Platform Engineering, a GE HealthCare software team
UDP	User Datagram Protocol
VRE	Volume Reconstruction Engine
VRF	Volume Reconstruction Receive Filter

* Printed copies are uncontrolled unless otherwise identified *
Before using this document, consult MyWorkshop for the latest revision.

1.5. Ownership

This document is owned by the Cadence-2024 Software Enterprise Architect.

1.6. Authorization

Refer to MyWorkshop for this document's approval, effectivity, and review information.

1.7. History

Table 1-2: Revision History

Rev	Description of Change	Reason for Change (or Change Control Number)	Author
1	▪ Cloned from Cad-2023H1 SWADD. ▪ Updated DOC references for 2023H2.	▪ Initial ▪ For 2023H2 work	A Byshenk (212070039)
2	▪ Updated Project to Cad-2024. ▪ Clarified coverage of MR-only, not PETMR. ▪ Updated Program name to Capri-B. ▪ Updated Program name to Tang-B. ▪ Updated DOC references ▪ Added Terms to Glossary ▪ Clarified MNI Overview. ▪ Clarified PSS Overview. ▪ Added Imaging Platform Overview. ▪ Clarified HOST description. ▪ Added HOST Gen8. ▪ Clarified VRE description. ▪ Clarified Imaging Platform descript. ▪ Added Boost Library. ▪ Clarified MR Recon structure. ▪ Added MR Recon Acquisition section and clarified content. ▪ Added MR Recon Image Generation section and clarified content. ▪ Added MR Recon Prescan section and clarified content. ▪ Added MR Recon Anatomy Recognition section. ▪ Added MR Recon Real-Time Processing section and clarified. ▪ Added MR Recon Raw Data Access section and clarified content. ▪ Added ICE to MR FW Receive Chain. ▪ Added MR FW Cabinet Monitor. ▪ Clarified MR Service Software structure.	▪ Nomenclature change ▪ For clarity ▪ Capri split to phases A and B. ▪ Tang split to phases A and B. ▪ For M3 ▪ For clarity ▪ For consistency with CX SWADD. ▪ For consistency with CX SWADD. ▪ Coverage ▪ For consistency with CX SWADD. ▪ New ▪ For consistency with CX SWADD. ▪ For consistency with CX SWADD. ▪ For consistency with CX SWADD. ▪ For consistency with CX SWADD. ▪ For consistency with CX SWADD. ▪ For consistency with CX SWADD. ▪ For consistency with CX SWADD. ▪ For consistency with CX SWADD. ▪ For consistency with CX SWADD. ▪ New HW ▪ Coverage ▪ For consistency with CX SWADD.	A Byshenk (212070039)

* Printed copies are uncontrolled unless otherwise identified *
Before using this document, consult MyWorkshop for the latest revision.

Released

Rev	Description of Change	Reason for Change (or Change Control Number)	Author
	▪ Added MR Service Software, Software Tools section. ▪ Added MR Service Software, System Tools section. ▪ Added MR Service Software, Service Tools section. ▪ Clarified Function Shut Down System. ▪ Clarified Function Error Handling and Recovery. ▪ Clarified Safety Segregation of critical data. ▪ Clarified Safety Segregation, Patient Identification Protection. ▪ Added IQE Client content to: Host Software Structure (4.1) User Interfaces (5.1) External Interfaces (5.2) ▪ Added references to IQ Engine SWADD. ▪ Removed material applicable to DRF, VRF, MGD. ▪ Updated Host Platform SSDD reference.	▪ For consistency with CX SWADD. ▪ For consistency with CX SWADD. ▪ For consistency with CX SWADD. ▪ For consistency with CX SWADD. ▪ For consistency with CX SWADD. ▪ For consistency with CX SWADD. ▪ For consistency with CX SWADD. ▪ New functionality ▪ Relevant information ▪ Obsolete per FW Arch ▪ New for MR Cad-2024	

*** Printed copies are uncontrolled unless otherwise identified ***
Before using this document, consult MyWorkshop for the latest revision.

2. SOFTWARE OVERVIEW

2.1. Context

This section describes the high-level relationships of the Software to the Scanner in which it operates.

The Software context is depicted in Figure 2-1.

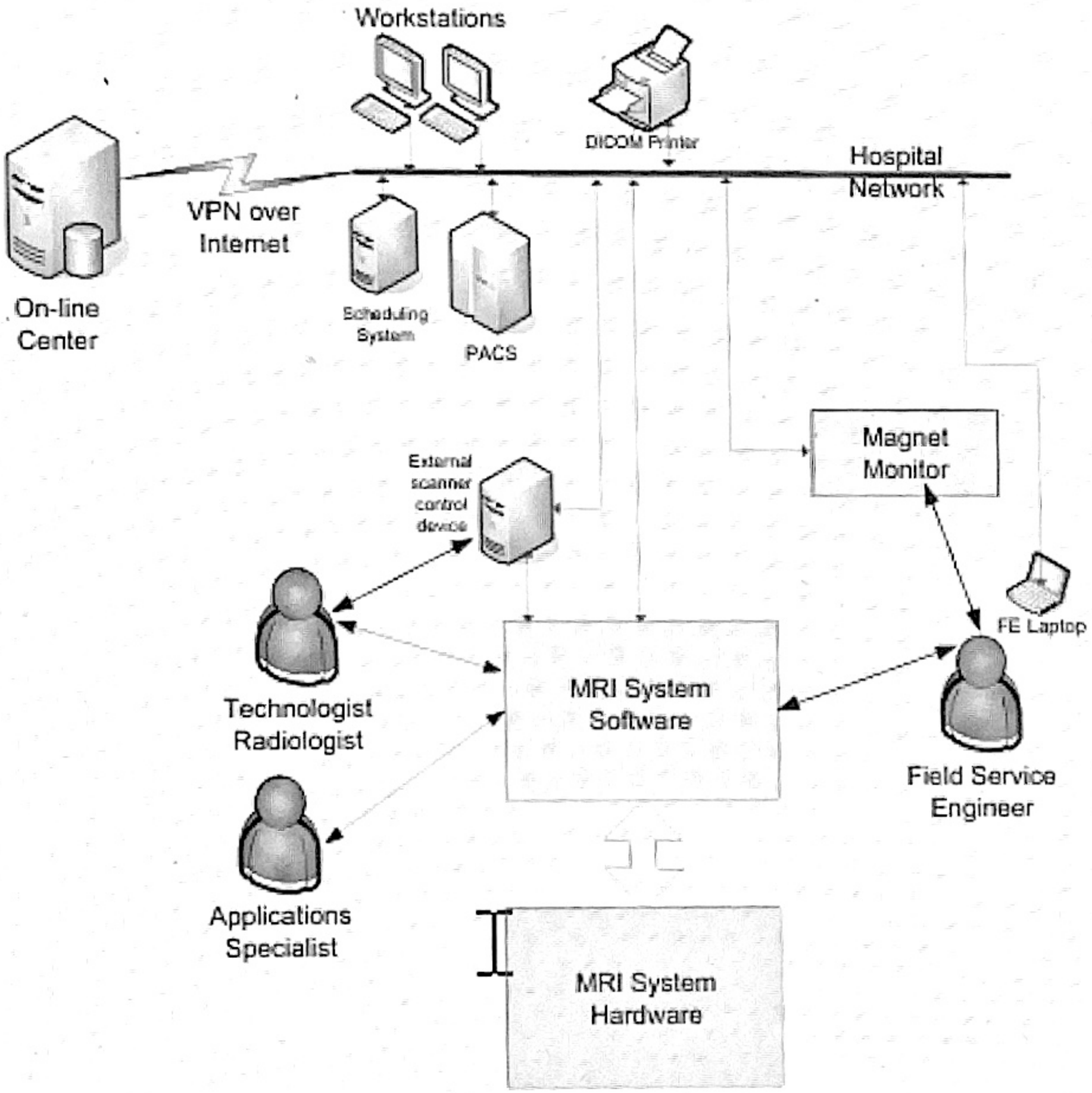


Figure 2-1: Context Overview

The primary users of the Scanner are:

Role	Activities
Technologist	Performs clinical scans
Reading Physician	Interprets images generated by the Scanner

*** Printed copies are uncontrolled unless otherwise identified ***
Before using this document, consult MyWorkshop for the latest revision.

Role	Activities
Field Service Engineer	Installs, calibrates, and maintains the Scanner
Application Specialist	Provides technical guidance to Technologists and Reading Physicians on the proper operation of the Scanner
Manufacturing Technician	Ensures parts are working before shipment

The Scanner connects to:

External System	Function
Scheduling System	Provides work orders to the Scanner Work orders include patient information and scheduled exams.
PACS	Provides a repository for images aka Picture Archiving and Communication System
DICOM Printer	Generates hardcopies of images
Workstation	Provides off-Scanner review and processing of images
FE Laptop	Aids the field service engineer during service operations
Online Center	Provides remote service and training support for the Scanner
Scanner Control Device	Allows external control of the Scanner Used by applications such as Focused Ultrasound treatment.

2.2. Uses

The general uses supported by the Software include:

Primary Uses

- Set Up Scans (Prescribe, Prepare)
- Acquire Data (Prescan, Acquire)
- Reconstruct Images
- Post-Process Images
- View Images

Secondary Uses

- Manage Scans
- Manage Exams
- Manage Patients
- Manage Protocols
- Manage Tracers
- Manage Images
  includes Exchange Images with DICOM devices and Archive-Restore
- Monitor System
  includes local checking, such as Log Event, View Events
  includes remote checking of devices, such as Magnet
- Qualify System
- Calibrate System
- Manage System
  includes Start Up, Shut Down, Reset, Log In, Log Out

* Printed copies are uncontrolled unless otherwise identified *
Before using this document, consult MyWorkshop for the latest revision.

includes Configure, such as Policies/Users/Groups, Licenses
includes Install, Uninstall, Upgrade

- Maintain System
- Diagnose System

2.3. Layering

Each software process is organized into layers that run on top of the hardware:

- System
- Service
- Application

The System layer provides the basic life cycle control of the Software as well as providing the device-specific hardware-software interfaces. The System layer also includes general functionality that is not proprietary to GE. Each computer within the system supports a form of an operating system at the base level.

Dependent on the System layer is the Service layer. The Services provide GE-proprietary functionality, typically acting on requests or responses originated by software modules at the Application or System layers.

Dependent on the Service and System layers, the Application layer provides data processing and interfaces with users (display and control) and components external to the Scanner.

*** Printed copies are uncontrolled unless otherwise identified ***
Before using this document, consult MyWorkshop for the latest revision.

The layers are represented graphically in Figure 2-2.

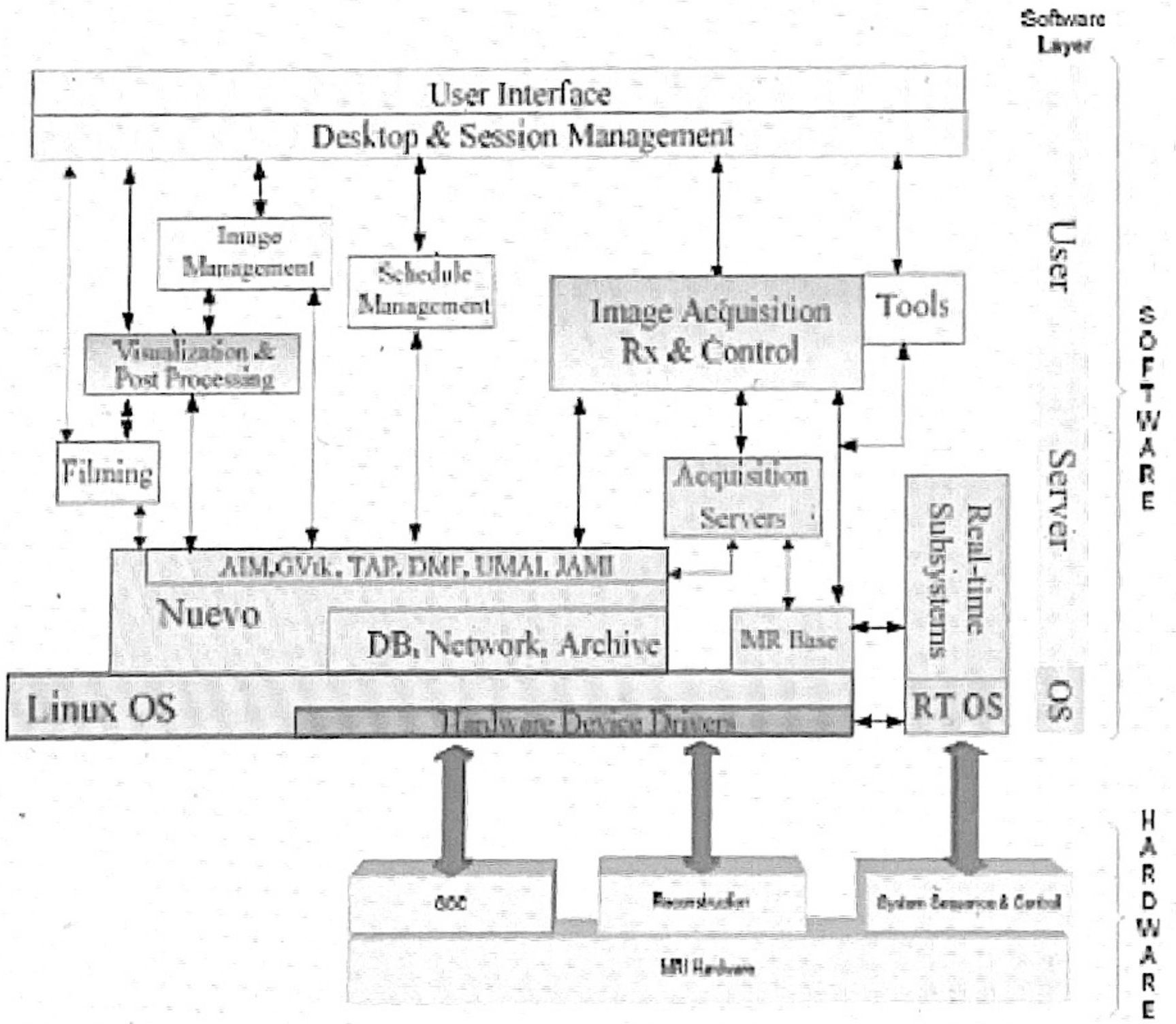


Figure 2-2: Software Layering

2.4. Subsystems

The Software is comprised of the following subsystems.

- Host Software (aka MR Host, aka Host)
- Host Platform
- Visualization Applications (aka Visualization, aka Viz)
- MR Reconstruction (aka MR Recon)
- Pulse Sequence Software (aka PSS)
- MR Firmware (aka MR FW)
- MR Service Software (aka Service SW, aka MR Tools)
- Imaging Platform

The subsystems are loosely aligned with the hardware design of the Scanner as shown in Figure 2-1: Context Overview.

The subsystem deployment is shown in Figure 2-3.

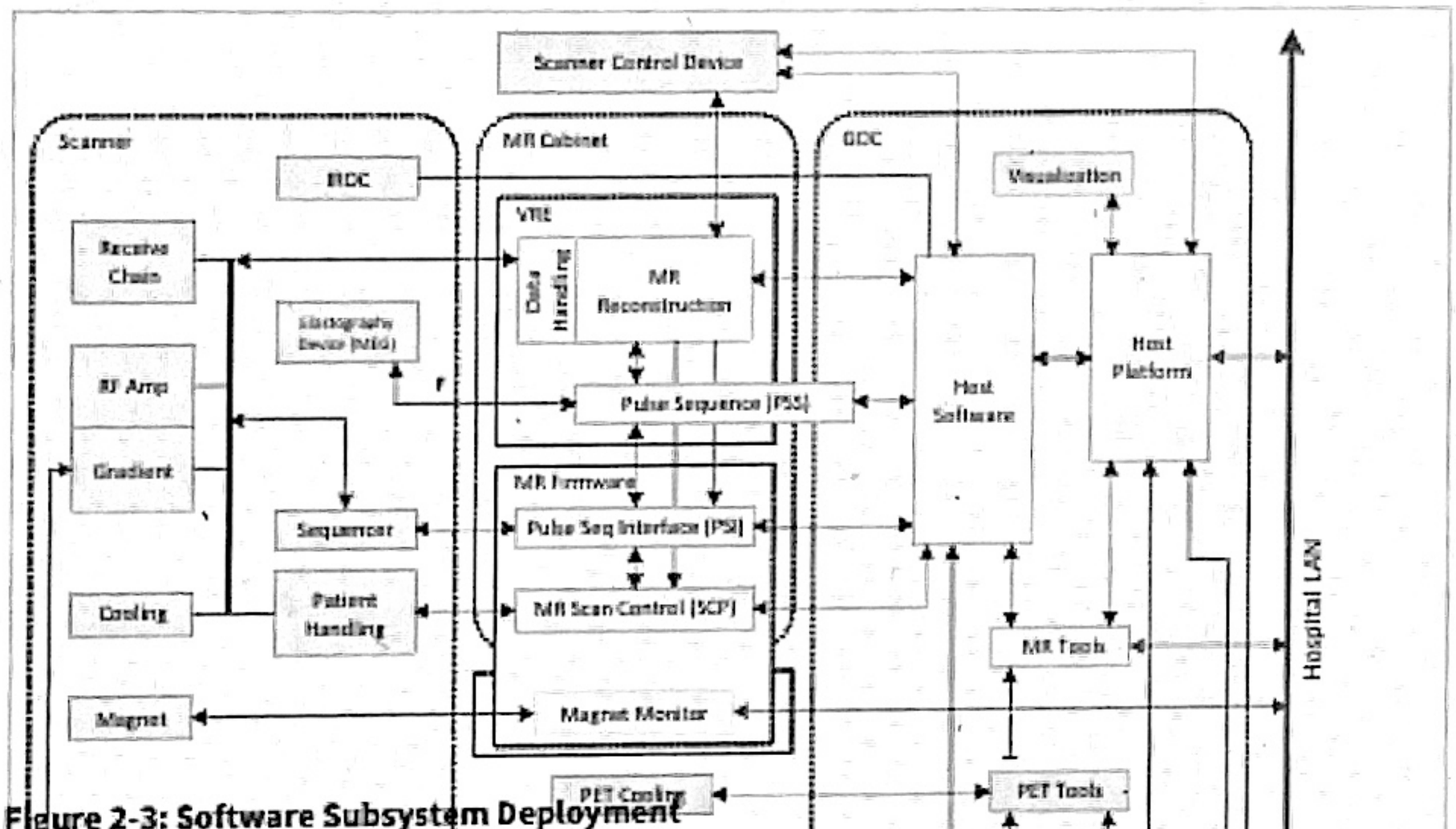


Figure 2-3: Software Subsystem Deployment

Each clinical exam is performed by coordinating interactions between the software subsystems. The workflow for each clinical exam begins with Host and Host Platform loading a protocol and instantiating the collection of Tasks and associated Applications identified in the protocol. Each Application is used to control scanning and/or post-processing. For each scanning task, Host launches a PSD and provides a user interface allowing the technologist to prescribe parameters appropriate for the Task. Host facilitates the workflow of Tasks and Applications by sequencing scan operations using scan controls such as download, start, pause, resume, and stop. Performing a scan operation also involves PSD and MR Firmware running on the scan control machines and MR Recon running on VRE. MR Recon delivers images to Host's image handling components. Post-processing of images is handled by MR Visualization Applications running on the host computer.

The following subsections provide a summary of the specific functions performed by each software subsystem.

2.4.1. Host Software

The Host Software subsystem (aka MR Host, aka Host) provides interfaces (Input and output) and fundamental functionality for the Scanner's scan control, covering:

- Prescribe (e.g. patient position & orientation, physio gating, coil selection, geometry, sequences, imaging options, GRx, Coil data, Auto-Coil, Auto-Prescribe, Brainwave RXM)
- Acquire (e.g. prescan, spectroscopy, navigator, fMRI, RTIA, RTSAR, Auto-Start, Auto-Scan, Brainwave PXE & RT, external control – EXSI, LAIS, SVAT)
- Construct (e.g. header generation, image installation, post-processing, ExpressSpine)
- Monitor (e.g. tasks, power, Auto-View).

* Printed copies are uncontrolled unless otherwise identified *
Before using this document, consult MyWorkshop for the latest revision.

Some Host applications that incorporate third-party components (e.g. Medical Numerics RXM/PXE/RT) use adapters to interface with them. An example interaction is shown in Figure 2-4.

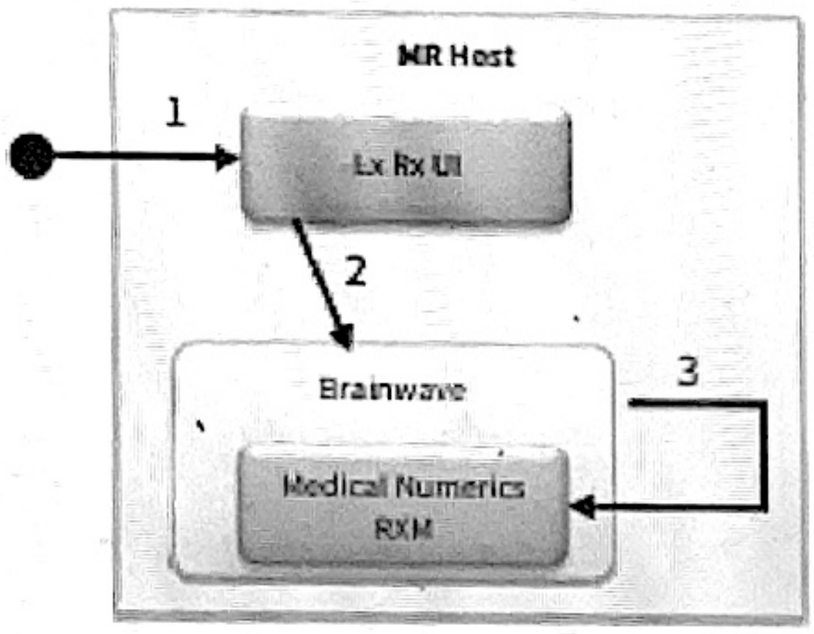


Figure 2-4: Host Adapted 3P Application Interaction

Host runs on HOST.

2.4.2. Host Platform

The Host Platform subsystem provides much of the user interface (input and output) for the Scanner's non-scan functionalities, covering:

- DICOM networking (which includes image transport, DICOM print, worklist receive and DICOM performed procedure step management)
- Patient Schedule (which includes the modality worklist user interface)
- DICOM Image Database
- User Interface Management
- System Startup
- Logging / Viewing System Messages

Host Platform runs on HOST.

2.4.3. Visualization Applications

The Visualiation Applications subsystem (aka Visualization, aka Viz) provides both image display and image post processing capabilities for the Scanner. The post processing software allows the user to display original and derived images as well as enhance original images for display only. The user can also create derived images as well as synthesize derived images based on parameter settings plus a set of original or derived images.

Some Viz applications utilize the AW Creator Plug-In mechanism that provides a framework for selecting and accepting images, processing the content, producing new content, and installing the results to the Image DB. The mechanism also allows convenient access to various image manipulation functions included in Volume Share (VS). Viz applications that take advantage of this mechanism employ specialized Plug-Ins that extend the general AW Creator behavior.

* Printed copies are uncontrolled unless otherwise identified *
Before using this document, consult MyWorkshop for the latest revision.

Released

Some Viz applications that incorporate third-party components (e.g. Synthetic MRI Tool, Neosoft Flow Analysis) use adapters to interface with the third-party components. The logical organization is shown in Figure 2-5.

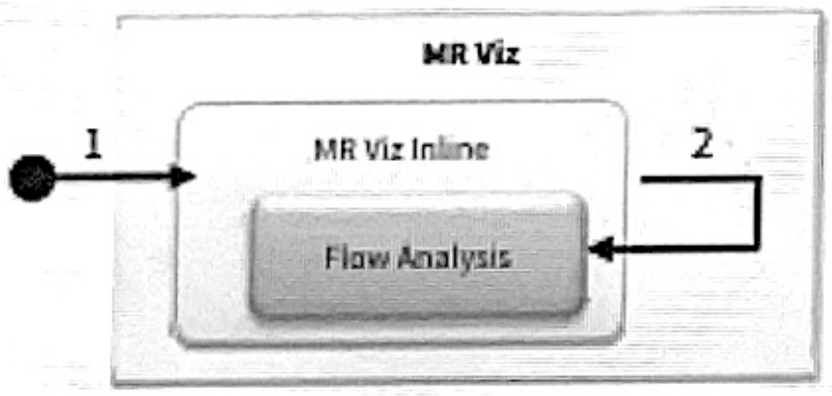


Figure 2-5: Viz Adapted 3P Application Interaction

Several Medical Numerics applications that are incorporated to Viz consist of a client application and one or more associated Processers managed by the MNI Platform, also provided by Medical Numerics. These applications use the client adapted into the Viz UI, and use the MNI Platform to execute the appropriate Processor, which will play out paradigms, control scans, or post-process images. An example interaction is shown in Figure 2-6.

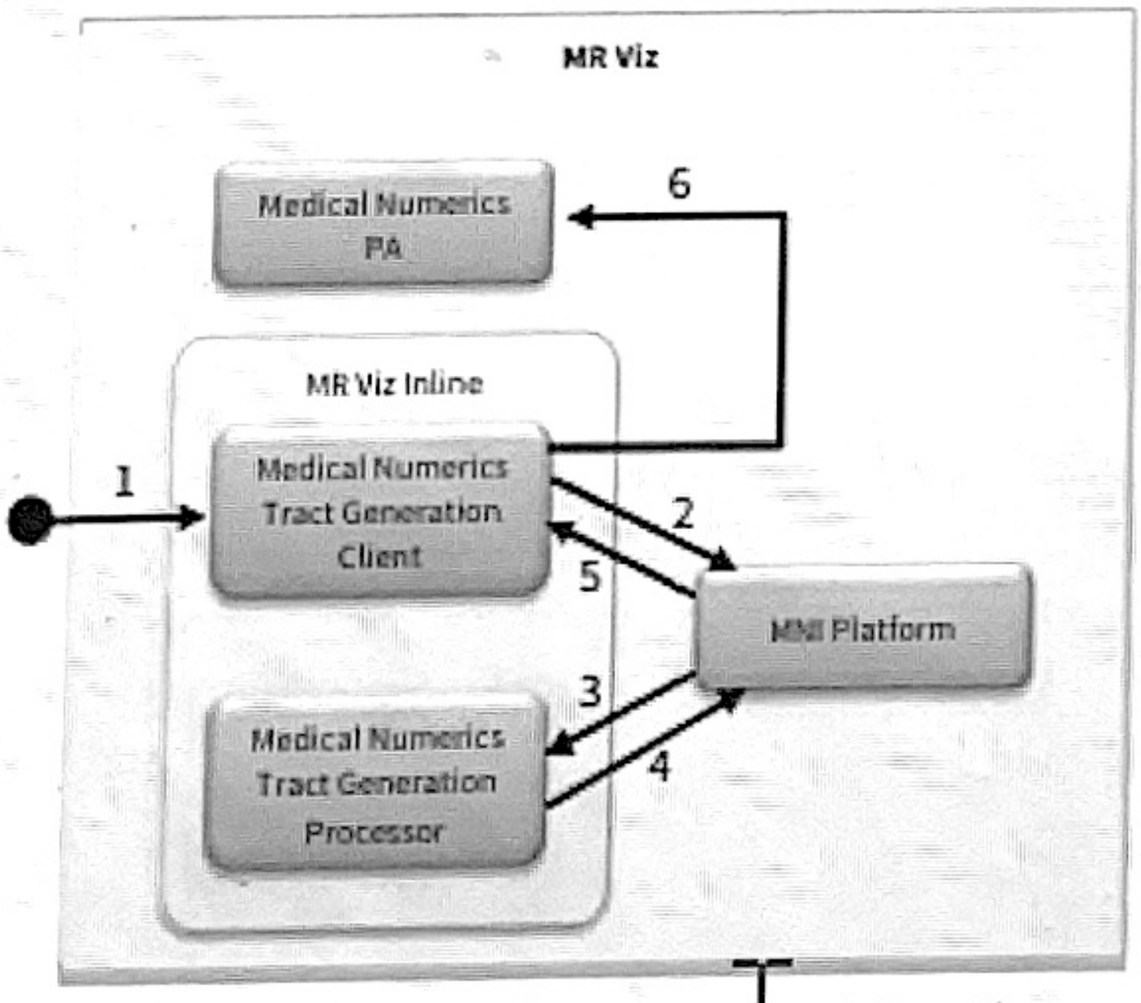


Figure 2-6: Viz MNI Application with MNI Platform Interaction

Viz runs on HOST.

2.4.4. MR Reconstruction

The MR Reconstruction subsystem (aka MR Recon) manages data collection from the RF Receive chain, filters the data to the desired bandwidth, organizes the data into K-space, performs the processing to transform the data to a useful form, and sends the results to the desired endpoint within Host. The results are typically images, but other outputs include spectral analysis, and/or Prescan parameters.

MR Recon also performs Anatomy Recognition activities and sends the results to Host. The results are typically Scan Parameters, including anatomy and geometry.

*** Printed copies are uncontrolled unless otherwise identified ***
Before using this document, consult MyWorkshop for the latest revision.

MR Recon typically runs on the Volume Reconstruction Engine (VRE) computers, but portions are developed to be location-independent, meaning they are containerized and can be run in any machine that can host the container and supports the necessary connectivity.

2.4.5. Pulse Sequence Software

MR applications use shaped RF and gradient pulses whose timing and operation are controlled by a pulse sequence timing diagram, a runtime description, and different levels of hardware control. A Pulse Sequence Description (PSD) is software used to set up and control the MR physics to meet the needs of an MR application. The PSD interacts with other software and hardware components tovalidate scanning task parameters and execute the task on the Scanner.

The Pulse Sequence Software (aka PSS) subsystem consists of PSDs and the components and functions used to execute those PSDs. The prescription portion of PSS is active for acquisition setup and runs on HOST. The acquisition target portion is active for controlling the run-time data acquisition and is executed by MR Firmware on ICE.

2.4.6. MR Firmware

The MR Firmware subsystem (aka MR FW) is the collection of software components that reside on several of the hardware components of the Scanner. There is a master interface for this control logic and hardware interface. For each of the hardware subsystems there are additional software items that reside on each of the smart peripherals within the hardware subsystem. In general, the software component for a particular hardware interface includes the software item in the master interface as well as the software item on the smart peripheral for that hardware.

The hardware subsystems controlled by MR Firmware are:

- Patient Handling
- Gradient
- Transmit Chain
- Receive Chain
- Cooling and Cabinet Monitor
- Magnet Monitor

MR Firmware components generate the timed control signals necessary to control MR imaging. This includes device initializations as well as data acquisition control. Acquisition control is performed under the direction of a PSD. A PSD drives control signals for the gradient output and RF output, as well as gating the RF receive windows. Data management information for the acquisition data is also generated, synchronized with the hardware controls. MR Firmware also receives control information from the physiological and external gating hardware. Those inputs provide gating/triggering for the PSD to synchronize data acquisition with external events.

MR Firmware has internal interfaces that communicate the command/control/status data between the individual software components and the associated hardware components. Additionally MR Firmware has a scan control component that manages the scanning states of

* Printed copies are uncontrolled unless otherwise identified *
Before using this document, consult MyWorkshop for the latest revision.
Released

the system and communicates the state changes to all of the software subsystems directly involved in the scanning process:

- MR Recon
- PSS
- Host

The Magnet Monitor allows FEs to remotely monitor liquid helium levels, magnet vessel pressure and magnet temperatures by detecting and reporting the information to the GE back office (on-line center). The Magnet Monitor runs on an independent hardware component.

2.4.7. MR Service Software

The MR Service Software subsystem (aka Service SW, aka MR Tools) provides functions to control activities of the following System uses:

- Monitor System
- Qualify System
- Calibrate System
- Configure System
- Install System
- Upgrade System
- Maintain System
- Diagnose System

This software runs on HOST.

2.4.8. Imaging Platform

The Imaging Platform subsystem provides applications and functions to support activities of the following System uses:

- Post-Process Images (Multi-Planar Reformat, DL Inference)
- View Images (Core Viewer)
- Manage Scans (Workflow Platform)
- Manage Exams (Modality Worklist, PPS)
- Manage Patients
- Manage Protocols
- Manage Images (Create, Read, Update, Delete, Exchange, Archive-Restore)
- Monitor System (Connect Remotely)
- Manage System (Log In, Log Out, Configure Policies/Users/Groups)

This software runs on HOST.

This software is provided to MR by GEHC groups outside MR and delivered via Coreload, VolumeShare, Edison Inference, and Cyber Package. CoreLoad is developed by GEHC's SWPE organization and is further described in the CoreLoad DHF [6]. VolumeShare is developed by GEHC's AW organization, and is further described in the VolumeShare DHF [7]. Edison Inference is developed by GEHC's STO. Cyber Package is developed by GEHC's SWPE organization and is further described in the CyberSecurity DHF [8].

The content of this Subsystem is used by multiple Imaging modalities.

*** Printed copies are uncontrolled unless otherwise identified ***
Before using this document, consult MyWorkshop for the latest revision.

2.5. Error Handling and Recovery

The error handling strategy for the Scanner consists of logging, automatic recovery, manual recovery and automatic shutdown.

Each subsystem is responsible to detect and log errors as well as pass status back to the component requesting the operation. When feasible, subsystems should attempt recovery (I.e. retry, auto restart, etc.).

When automatic recovery is not possible, operator intervention is required. This interaction can range from forcing a partial Scanner reset (see Section 6.2.6) to retrying the operation. For example:

- If there is insufficient disk space to begin image acquisition, the operator may remove unneeded images, freeing sufficient space to proceed.
- During prescription the Scanner will validate the operator input and provide suggested values to resolve incompatible parameter values.
- If the Cradle is not at the scan location, the operator may need to position the Cradle and retry the operation.
- A reconstruction or firmware fault may require a subsystem reset. The operator is informed of the issue (the specific fault condition is logged) and is requested to perform a reset of the acquisition hardware (aka TPS Reset).

In rare cases a catastrophic system fault occurs, which causes an automatic shutdown of the Scanner. The Scanner will attempt to log the fault to facilitate subsequent repair and/or remediation.

2.6. Risk Control

All system hazards are identified in the MR RAC [11]. The following critical hazards are associated with the Software.

- RAC ID 2.200.30 - Excessive SAR (RF)
- RAC ID 10.100.130 - Orientation or annotation of images (primarily relied upon)
- RAC ID 10.100.142 - Sagittal image ordering (primarily relied upon)
- RAC ID 10.100.170 - Data/Patient mismatch

The Software may contribute to additional minor and serious hazards. All hazards associated with software items are recorded in the CMT [13] and software FMEAs [20], [24], [22], [23], [24], [26], [27]. Safety assessments and classifications of software items are recorded in the SW Structure and Safety Classification [19].

Because the Software may contribute to critical hazards, it is considered Safety Class C. The architecture has been designed to segregate the software components related to the critical hazards. This design allows additional focus on the Safety Class C components within the Software. For information on Safety Segregation, see section 8 of this document.

The technical design review process coupled with mandatory code reviews by the subject matter experts ensures that the architectural segregation is both effective and maintained.

Software safety requirements fulfill mitigations in the CMT [13].

*** Printed copies are uncontrolled unless otherwise identified ***
Before using this document, consult MyWorkshop for the latest revision.

Released

2.7. Legacy Reuse

The architecture described in this document reuses many components developed for legacy programs; however, explicit reuse is driven by each subsystem, dependent on features (HW & SW) associated with the products supported by this Software. Specifics on reusability are covered within the subsystem SSDDs.

Across subsystems, one primary set of components designed for reusability provides the inter-process communication capabilities for the Scanner. These communication channels are implemented using the following basic communications transports:

- TCP/IP
- Shared Memory
- Files
- POSIX Message Queues
- X-Atoms
- RPC

These low level transports have been abstracted into the following high level communication APIs:

- MrMail/CM – custom packet based communication protocol supporting both Linux and VxWorks tasks supporting named lookup for endpoints. Built on-top of TCP/IP and RPC. MrMail is the high level API on the host; CM is the high level API on the VRE's and VxWorks boards.
- CORBA – common object request broker
- GEOCOMM – GE Object communication model providing cross platform object communication supporting both C++ and Java.
- Shared Memory – implemented via SVR4 shared memory or memory mapped files
- Message Queues – POSIX message queues
- CUP/IPC – inter-process communication protocol layered on top of TCP/IP, but with a simpler API.
- Memory Server – used to transport large blocks of data between processes residing on separate compute nodes (generally images but can be any large data block)
- SVAT Interface for automated scanner control
- MX framework communication interfaces: Mx event, Mx spider bus.
- AIM/UMAI – framework for applications communications with image management platform

In general, each of the above communication APIs is used for point-to-point communications.

Within the post processing area, the underlying computational logic has been partitioned from the user interface layer allowing the same processing library to provide both inline post processing as well as user directed (post processing session) based post processing.

2.8. Configurability

Aspects of this architecture are common across all Scanner Variants specified for the applicable releases (see section 1 of this document), including Variants consisting of differing

* Printed copies are uncontrolled unless otherwise identified *
Before using this document, consult MyWorkshop for the latest revision.

Released

hardware components and Variants supporting differing features and functions. The Software includes components supporting all the applicable Variants.

Components of the Software may be enabled/disabled based on one or more of the following factors:

- Automatic detection of specific hardware
- Selections made during installation or configuration
- Presence/absence of specific licenses

In most cases when it is possible, hardware-dependent Software components are enabled/disabled based on automatically detecting the presence of the associated hardware. In other cases, Software components that are dependent on hardware are enabled/disabled based on the installation or configuration selections. The mechanisms used to control the availability of specific hardware-dependent components are described in the component-level design documents.

Components of the Software which are independent of specific hardware, but are intended to vary based on the product or product line configurations are typically enabled/disabled based on selections made during installation. The mechanisms used to control the availability of these components are described in the component-level design documents.

Access to some features and functions is controlled by the presence/absense of licenses. This type of control is commonly referred to as an Option. Some features that are controlled as Options have licenses sold separately from base Systems, while the licenses for some such features are provided with the base Systems. The component-level design documents describe features that are controlled using Options.

2.9. Off-the-Shelf Components

The Software uses many Off-the-Shelf (OTS) components – that is, components from third parties that are unmodified for use in the Software. Nearly all the OTS components in the Software are used widely in many industries, benefitting the Software by external substantial runtime that discovers anomalies and significant debugging to understand and address any issues.

2.10. Extensibility

This architecture uses substantial subsystem (domain) segregation (see section 2.4), allowing extensions within each subsystem without directly affecting other subsystems. This architecture also separates business components from the Operating Environment (see section 2.3), allowing for extensions in the Operating System, Engines, and Platforms without disrupting the business components.

* Printed copies are uncontrolled unless otherwise identified *
Before using this document, consult MyWorkshop for the latest revision.

Released

3. ENVIRONMENT

3.1. Hardware

Figure 3-1 shows the primary Scanner hardware components that interface with the Software.

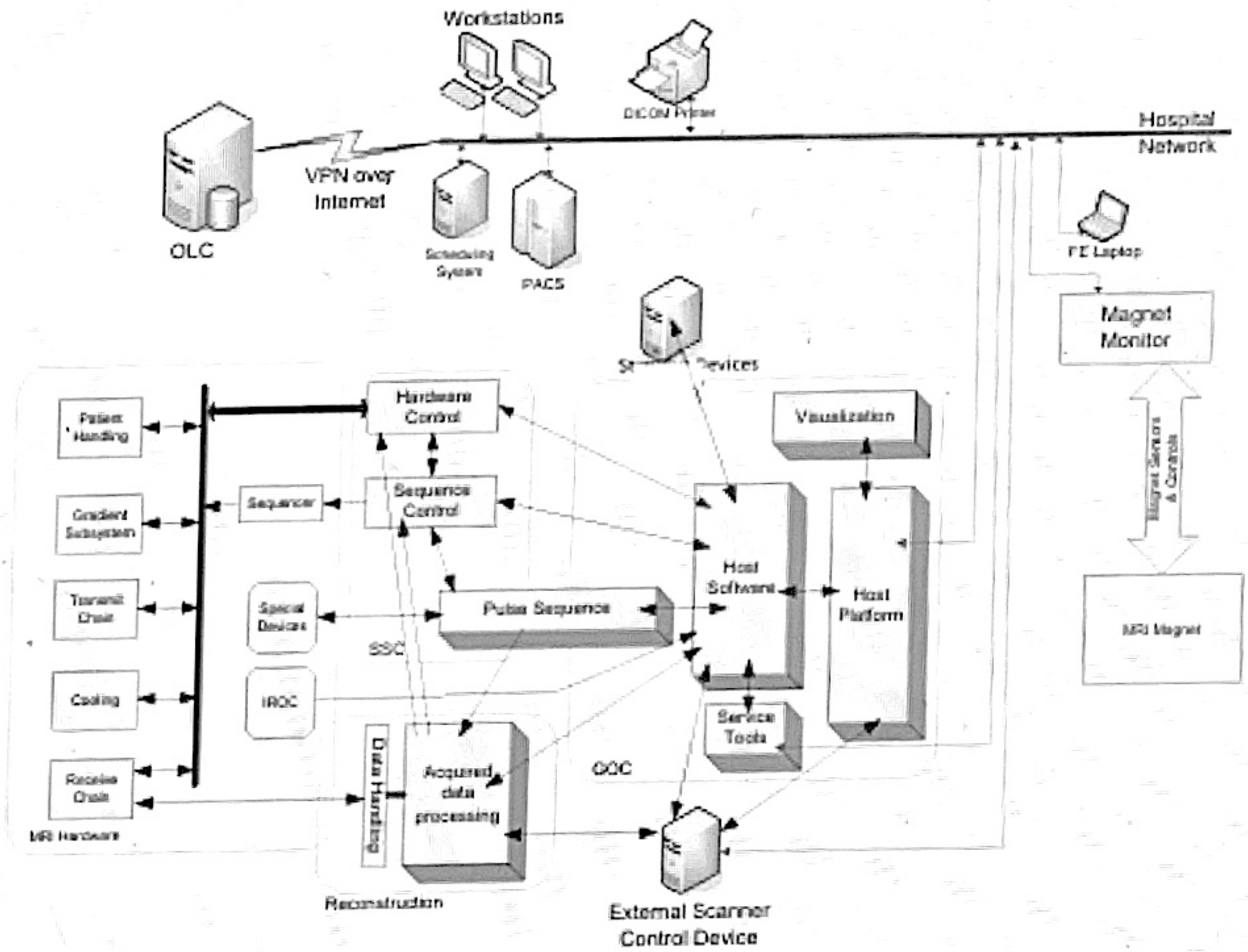


Figure 3-1: Hardware Organization

There are multiple computers used in the Scanner. The Host Computer (HOST) is housed in the Global Operator Console (GOC), a cabinet typically sited in the Operator Control Room. The operator interacts with HOST via the GOC monitor, keyboard, mouse, and SCIM (the MR proprietary pushbutton/intercom module). The Volume Reconstruction Engine (VRE) computer is headless, rack-mounted in the MR Cabinet located in the Equipment Room. The VRE is connected to the MR receive chain hardware.

The Scanner utilizes a dedicated ethernet subnet connecting the GOC to the MR Cabinet (the MR subnet). The HOST is connected to the dedicated subnet (and typically to a network provided by the installed site). Several devices in the MR Cabinet are connected to the MR subnet, including the VRE, the embedded Scan and Sequence Control computers, and other hardware peripherals.

The Software supports several versions of the compute hardware, including some models that exist only at Installed Base (IB) sites. The compute hardware items annotated with * identify those supported in forward production.

The HOST is an x86-based multi-core workstation running Linux, including display GPUs. The Software supports HOST models:

Name	Vendor Product	Processor(s)	Volatile Memory	Persistent Memory
Gen7	DELL T5820XL	1 × Intel Xeon, Skylake 4-core @3.6 GHz 2 × Nvidia P620	64 GB 2666 MT/s	2 × 512 GB SSD
Gen8*	HP Z4G5	1 × Intel Xeon, Sapphire Rapids 6-core @2.1/4.2 GHz 2 × Nvidia T1000	64 GB 4400 MT/s	2 × 1 TB SSD (NVMe)

The VRE consists of an Image Compute Node (ICN). An ICN is an x86-based multi-core, multi-CPU server running Linux, optionally including general purpose GPUs. The Software supports ICN models:

Name	Vendor Product	Processor(s)	Volatile Memory	Persistent Memory
Gen6-Value	Dell R630XL	2 × Intel Xeon, Sandy Bridge, 6-core	48 GB	3 × 400 GB SSD
Gen6-Perf	Dell R630XL	2 × Intel Xeon, Sandy Bridge, 12-core	96 GB	3 × 400 GB SSD
Gen6-Adv	Dell R630XL	2 × Intel Xeon, Sandy Bridge, 14-core	192 GB	3 × 400 GB SSD
Gen7-Value	Dell R640	2 × Intel Xeon, Skylake, 8-core @2.1 GHz	64 GB	1 × 480 GB SSD
Gen7-Perf	Dell R640	2 × Intel Xeon, Skylake, 12-core @2.3 GHz	128 GB	2 × 480 GB SSD
Gen7-Adv	Dell R640	2 × Intel Xeon, Skylake, 16-core @2.1GHz	192 GB	3 × 480 GB SSD
Gen7-DL-Perf	Dell R640	2 × Intel Xeon, Cascade Lake, 12-core @2.2 GHz 1 × Nvidia Tesla T4	128 GB	2 × 480 GB SSD
Gen7-DL-Adv	Dell R640	2 × Intel Xeon, Cascade Lake, 16-core @2.3GHz 1 × Nvidia Tesla T4	160 GB	3 × 480 GB SSD
Gen8*	Dell SIP-501-G1	2 × Intel Xeon, Sapphire Rapids 12-core @2.0/3.9 GHz	64 GB 4000 MT/s	1 × 1.6 TB SSD (NVMe)
Gen8-DL*	Dell SIP-503-G1	2 × Intel Xeon, Sapphire Rapids 12-core @2.0/3.9 GHz 2 x Nvidia 4000 Ada	192 GB 4000 MT/s	1 × 1.6 TB SSD (NVMe)

Software installation on these computers is provided by Service SW (section 4.7).
HOST boot operations are described in the Start Up System function (section 6.2.4).
MR Recon manages the VRE boot and startup configuration (section 4.4).

3.2. Operating Software

Operating Software for a machine provides applications running on the machine to interact with the machine's physical components (e.g. processors, memory/storage, user interface devices, networks, power) by providing logical components (e.g. processes, threads, files, streams, viewports, events) that are accessible through well-defined contracts (APIs and specified behaviors). Operating Software includes software items commonly known as Operating Systems, Virtual Machines & Containers, and Runtime Engines.

3.2.1. Linux

Linux is a general-purpose OS used on the HOST and VRE machines. The Software uses SUSE Linux Enterprise Server 15 (SLES), an implementation of Linux developed by SUSE and

*** Printed copies are uncontrolled unless otherwise identified ***
Before using this document, consult MyWorkshop for the latest revision.

provided to MR via the GEHC Software Central team. SLES is built on the open source Linux kernel. In addition to the functionalities described above for Operating Software, SLES provides containers and container management, provisioning (install/patch/configure facilities), security (resource access control, encryption), and the databases used for images and coil information. This software is further described in the Host Operating System SOUPRS [28] and the VRE OS SW SOUPRS [29].

3.2.2. VxWorks

VxWorks is an embedded real-time OS used on the Scanner sequence and control computer boards. VxWorks is a commercial product from Wind River. This OS has a small footprint and is designed to meet critical deterministic timing requirements of embedded real-time systems. This software is further described in the VxWorks OS SOUPRS [30].

3.2.3. eCos

eCos is an open source, real-time OS used to run some Gradient Firmware. It is a commercial product from eCosCentric, an outgrowth of Red Hat. This software is further described in the NIOS II SOUPRS [31].

3.2.4. MicroC/OS-II

MicroC/OS-II is a real-time OS for FPGAs used to run the Scan Room Interface (SRI) and some Gradient Firmware. It is a commercial product from Altera, a subsidiary of Intel. This software is further described in the NIOS II SOUPRS [31].

3.2.5. ACE

The ACE package is used by Host, MR Recon, PSS, and MR Firmware to provide an operating system abstraction layer. The ACE software is covered in the ACE TAO OpenORB SOUPRS [31] for Host, the Recon Libraries SOUPRS [36] for MR Recon, and the Firmware ACE SOUPRS [33] for PSS and MR Firmware.

3.2.6. Java

The Java Runtime Environment (JRE) is used to execute Java-based code in Host, Host Platform, Viz, Service SW. The Software uses Amazon's Corretto 64-bit implementation of the OpenJDK (open source) JRE and OpenLogic's 32-bit implementation of the OpenJDK JRE. This software is further described in the Host Software SOUPRS [30].

3.2.7. Python

The CPython engine is used to execute Python-based test code. The Python code may launch processes that are part of the product distribution or interact with running processes.

* Printed copies are uncontrolled unless otherwise identified *
Before using this document, consult MyWorkshop for the latest revision.